WORKED SOLUTION

STEP II 1998, Question 9

Mechanics • Statics of Frameworks

STEP II 1998, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s2-q9

The key idea

The framework is light, so only two external forces act on it: the support at A and the weight at C. Two forces in equilibrium must be equal, opposite and along the same line, and that settles both the support force and how the hexagon hangs. After that it is the method of joints: start at a joint where only two rods meet, resolve, and work round the hexagon one joint at a time, always moving to a joint with at most two unknown forces.

Solution

The supporting force

Model the hexagon's sides and the three diagonal rods as light rods, smoothly pinned at the six vertices, and write W for the weight of the 1 kg mass, so $W = g \approx 9.8\,\text{N}$. The rods are light, so the only external forces on the whole framework are the force $\mathbf{R}$ from the support at A and the weight W acting vertically downwards at C.

? **Ask yourself** Can the framework hang at any angle we like?

No. For the framework as a whole, the forces must sum to zero and have zero total moment. Taking moments about A, the support force has no moment, so the weight at C must also have zero moment about A. That happens only if C is directly below A. So the framework hangs with the short diagonal AC vertical, and resolving gives

$$\mathbf{R} = W \text{ vertically upwards}, \qquad |\mathbf{R}| = g \approx 9.8\ \text{N},$$

the weight of 1 kg acting along CA.

The forces in the rods

Take side length 1 (the forces don't depend on it). With AC vertical, B lies at the mid-height of AC, on one side of it, and the rest of the hexagon lies on the other side:

F level with A, E level with B, D level with C. So AF, BE and CD are horizontal, the sides AB, BC, DE, EF make 60° with the horizontal, and the short diagonals BD, BF make 30° with the horizontal.

Let T_{XY} be the tension in rod XY; at each end it pulls the joint towards the other end. A negative tension means the rod is in thrust.

Ask yourself There are nine unknown rod forces. Where do we start?

At a joint where only two rods meet, because two equations (horizontal and vertical) can then be solved at once. Joint C has only CB and CD, and it carries the load.

Joint C. Rod CB points 60° above the horizontal and CD is horizontal, away from B's side. Resolving,

$$\begin{aligned} \text{vertically:} \quad & T_{BC}\sin 60^\circ = W \;\Rightarrow\; T_{BC} = \tfrac{2W}{\sqrt{3}}, \\ \text{horizontally:} \quad & T_{CD} = T_{BC}\cos 60^\circ = \tfrac{W}{\sqrt{3}}. \end{aligned}$$

Joint D. Now only DE and DB are unknown. From D, rod DC points horizontally towards B's side, DE points 60° above the horizontal away from it, and DB points 30° above the horizontal towards it. Resolving,

$$\begin{aligned} \text{vertically:} \quad & T_{DE}\sin 60^\circ + T_{BD}\sin 30^\circ = 0 \;\Rightarrow\; T_{BD} = -\sqrt{3}\,T_{DE}, \\ \text{horizontally:} \quad & T_{CD} - T_{DE}\cos 60^\circ + T_{BD}\cos 30^\circ = 0 \;\Rightarrow\; T_{CD} - \tfrac{1}{2}T_{DE} - \tfrac{3}{2}T_{DE} = 0, \end{aligned}$$

so $T_{DE} = \frac{1}{2}T_{CD} = \frac{W}{2\sqrt{3}}$ and $T_{BD} = -\frac{W}{2}$.

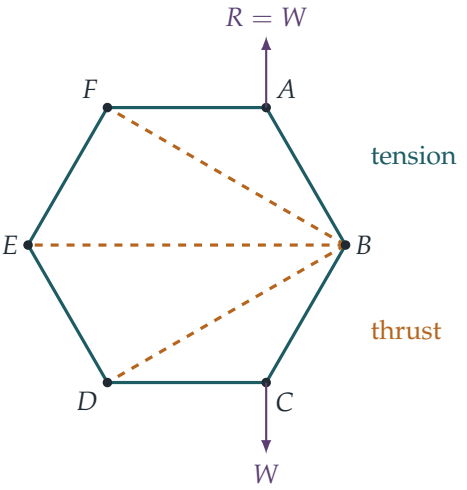
Joint E. Rod ED points 60° below the horizontal, EF points 60° above it, and EB is horizontal, all towards B's side. Resolving vertically gives $T_{EF} = T_{DE} = \frac{W}{2\sqrt{3}}$. Resolving horizontally,

$$\tfrac{1}{2}T_{DE} + \tfrac{1}{2}T_{EF} + T_{BE} = 0 \;\Rightarrow\; T_{BE} = -\tfrac{W}{2\sqrt{3}}.$$

Joint F. This is the mirror image of joint D in the line BE. Resolving vertically, $T_{BF} = -\sqrt{3}\,T_{EF} = -\frac{W}{2}$; horizontally, $T_{FA} = 2T_{EF} = \frac{W}{\sqrt{3}}$.

Joint A. Horizontally, $T_{AB}\cos 60^\circ = T_{FA}$, so $T_{AB} = \frac{2W}{\sqrt{3}}$. Vertically, $R = T_{AB}\sin 60^\circ = W$, agreeing with the first part.

Joint B gives two more equations, but they hold automatically: the framework as a whole is already in equilibrium. Checking them is a good test of the arithmetic.



Summarising, with $W = g$:

- tension in all six sides: AB and BC carry $\frac{2W}{\sqrt{3}}$, CD and FA carry $\frac{W}{\sqrt{3}}$, DE and EF carry $\frac{W}{2\sqrt{3}}$;
- thrust in all three inner rods: BD and BF carry $\frac{W}{2}$, and BE carries $\frac{W}{2\sqrt{3}}$.

So the force in BE is a thrust of magnitude

$$\frac{g}{2\sqrt{3}} = \frac{\sqrt{3}\,g}{6} \approx 2.83 \text{ N}.$$

Checks

- Symmetry: reflecting in the line BE swaps $A \leftrightarrow C$ and $F \leftrightarrow D$, and turns the upward force W at A into the downward force W at C. So the loading is symmetric, and the answers should be too: $T_{AB} = T_{BC}$, $T_{FA} = T_{CD}$, $T_{EF} = T_{DE}$, $T_{BF} = T_{BD}$. They are.
- Joint B: horizontally, the two sides give $2 \cdot \frac{2W}{\sqrt{3}} \cdot \frac{1}{2}$ one way, and BD, BF, BE give $2 \cdot \frac{W}{2} \cdot \frac{\sqrt{3}}{2} + \frac{W}{2\sqrt{3}} = \frac{2W}{\sqrt{3}}$ the other; vertically, AB and BC cancel, as do BD and BF.
- We solved all twelve joint equations numerically with $g = 9.8$: they are consistent, give $R = 9.8\,\text{N}$ upwards and $|T_{BE}| \approx 2.829\,\text{N}$ in thrust, and have no solution if AC is tilted.

Common slips

- Drawing the hexagon with a side or a long diagonal vertical. The hanging position is forced: C below A.

- Mixing up sign conventions between joints. Fix "tension pulls the joint towards the other end of the rod" once, and let negative answers tell you which rods are in thrust.

Beyond STEP

A framework with j joints is rigid and statically determinate when it has exactly $2j - 3$ rods: here $2 \cdot 6 - 3 = 9$, which is why the forces can be found by statics alone. With more rods the structure is "redundant" and you need the elasticity of the rods; this is the start of structural mechanics, and of the mathematics of rigidity of graphs. For the vector mechanics behind it, try Kleppner and Kolenkow, *An Introduction to Mechanics*.

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